THE UNIVERSITY OF QUEENSLAND
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This exam paper must not be removed from the venue

Venue _____
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Student Number |__|__|__|__|__|__|__|__|
Family Name _____
First Name _____

School of Information Technology and Electrical Engineering
Semester One Examinations, 2023
COMP4702 / COMP7703 Machine Learning

This paper is for St Lucia Campus students.

Examination Duration: 180 minutes

Planning Time: 10 minutes

Exam Conditions:

•This is a Closed Book examination - no written materials permitted
•Casio FX82 series or UQ approved and labelled calculator only
•During Planning Time - Students are encouraged to review and plan responses to the exam questions
•This examination paper will be released to the Library

Materials Permitted in the Exam Venue:
(No electronic aids are permitted e.g. laptops, phones)

None

Materials to be supplied to Students:
Additional exam materials (e.g. answer booklets, rough paper) will be provided upon request.

1 x Gradescope Bubble Sheet

Instructions to Students:
If you believe there is missing or incorrect information impacting your ability to answer any question, please state this when writing your answer.

Answer Part A in the Gradscope Bubble Sheet.

Answer Part B in the space provided in the question paper.

Total marks: 80.

For Examiner Use Only

Question	Mark

Total _____

Part A - Multiple choice questions (2 marks each). Record your answers on the bubble sheet provided.

Question 1.

Which of the following best describes supervised learning?

(a) A type of machine learning where the algorithm learns to optimize a reward function through trial and error.

(b) A type of machine learning where the algorithm learns to identify patterns and relationships in unlabeled data.

(c) A type of machine learning where the algorithm learns to make predictions based on input data and output labels.

(d) A type of machine learning where the algorithm learns to cluster data points based on their similarity.

Question 2.

Which of the following is NOT a commonly used activation function in neural networks?

(a) Logistic: $h(z) = \frac{1}{1+e^{-z}}$

(b) ReLU: $h(z) = \max(0, z)$

(c) Linear: $h(z) = z$

(d) sinh: $h(z) = \frac{1-e^{-2z}}{2e^{-x}}$

Question 3.

What is the purpose of a hold-out data set in machine learning?

(a) To train the model on new data that it hasn't seen before.

(b) To estimate the accuracy of the model on data that it has not been trained on.

(c) To fine-tune the model's hyperparameters and avoid overfitting.

(d) To estimate the accuracy of the model on data that it has already been trained on.

Question 4.

What does the "k" in the kNN algorithm represent?

(a) The number of training examples used for each prediction.

(b) The number of features in the dataset.

(c) The number of classes in the dataset.

(d) The regularisation term in the loss function.

Question 5.

Which of the following metrics is NOT suitable for evaluating the quality of a split in a decision tree for classification?

(a) Entropy.

(b) Gini index.

(c) Misclassification rate.

(d) Sensitivity.

Question 6.

Which of the following correctly describes stochastic gradient descent (SGD)?

(a) SGD computes the gradient of the objective function over the entire training set.

(b) SGD updates the model parameters using the gradient plus a random noise term.

(c) SGD computes the gradient of the objective function over a single training example.

(d) SGD computes the gradient of the objective function over a random subset of the validation set.

Question 7.

Regarding bagging, which of the following statements is *incorrect*?

(a) Bagging allows multiple models to be trained in parallel.

(b) Bagging uses bootstrap sampling to create multiple training sets.

(c) Bagging tends to increase bias when the size of the ensemble grows too large.

(d) Bagging tends to reduce the model error variance compared to a single base model.

Question 8.

Regarding the kernel function in support vector machines, which of the following statements is correct?

(a) The kernel function provides a measure of the similarity between any pair of data points.

(b) The kernel function reduces the dimensionality of the input space to increase the separation of classes in a classification problem.

(c) The kernel function allows a training problem to be formulated as an unconstrained convex optimisation problem.

(d) The kernel function controls the size of the margin.

Question 9.

What is the main difference between the k-means clustering algorithm and Gaussian mixture models (trained using the EM algorithm)?

(a) k-means clustering can handle data with missing values, while Gaussian mixture models cannot.

(b) k-means clustering uses a "hard" assignment of data points to clusters, whereas Gaussian mixture models use a "soft" probabilistic assignment.

(c) The EM algorithm is prone to getting stuck in local minima, whereas the k-means algorithm always converges to the global optimum.

(d) k-means clustering requires a technique for initialization whereas the EM algorithm does this automatically.

Question 10.

Which of the following statements about Principal Component Analysis (PCA) is *incorrect*?

(a) Given a dataset with p features, PCA can reduce the dimensionality of the data to a minimum of $\frac{p}{2}$ features.

(b) PCA performs dimensionality reduction via a linear projection of the data.

(c) PCA can be carried out by calculating a singular value decomposition of the centered data matrix.

(d) PCA can be carried out by calculating the eigenvectors and eigenvalues of the covariance matrix of the data.

(next question over)

Part B - Questions carry the number of marks indicated. Write your answers in the space provided after each part of each question.

Question 1.

Figure 1 shows a set of training data for a binary classification problem:

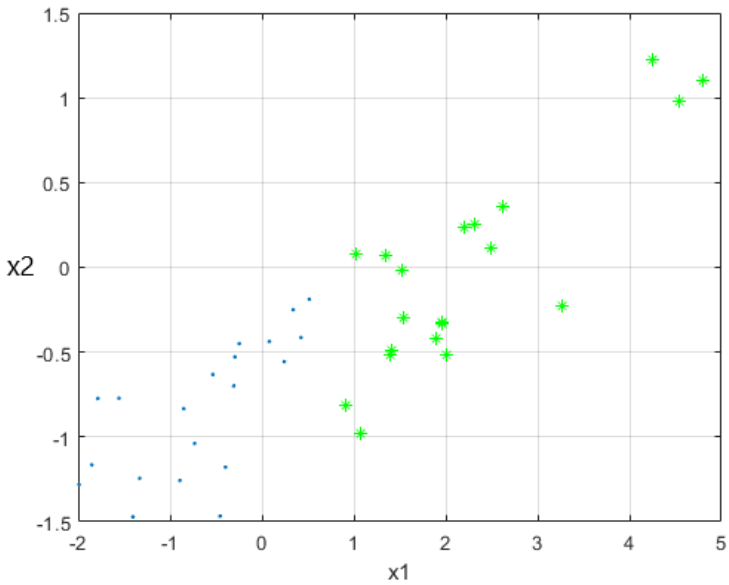

Figure 1

(a) Consider using a decision stump (i.e. a decision tree with depth = 1) to classify this data.

(i) From Figure 1, specify the trained model (approximately) in terms of an IF...THEN decision rule. (2 marks)

(ii) State the misclassification rate of the trained model on the training set and explain why. (2 marks)

(b) Consider performing logistic regression on the data from Figure 1. How many parameters would the trained linear regression model have? (2 marks)

(next question over)

Question 2.

Figure 2 is taken from a textbook by Chris Bishop:

Plot of the sum-of-squares test-set error for the polynomial data set versus the number of hidden units in the network, with 30 random starts for each network size, showing the effect of local minima. For each new start, the weight vector was initialized by sampling from an isotropic Gaussian distribution having a mean of zero and a variance of 10.

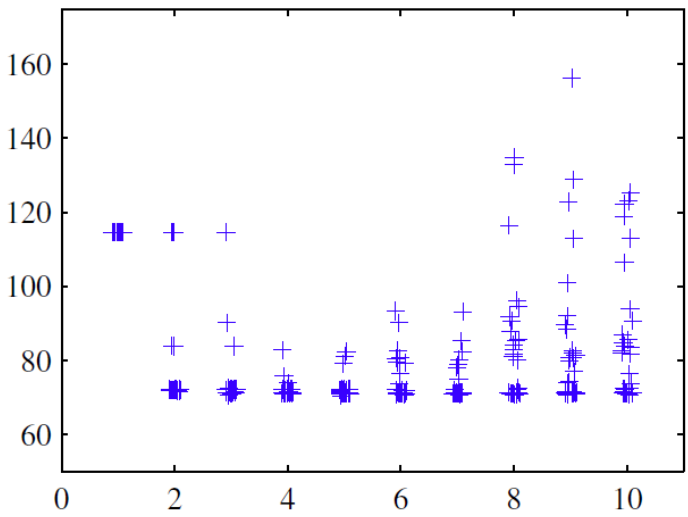

Figure 2

Note that a small amount of noise (jitter) has been applied to the horizontal positions of each data point, to make it easier to see that there are 30 results for each number of hidden units tested.

(a) Assume that the networks are all trained using gradient descent with a fixed learning rate. What does this graph tell us about the training process of a multi-layer perceptron? Discuss in a few sentences. (5 marks)

(question continues over)

(b) In his book, Bishop states that the lowest validation error obtained over all the training runs was for 8 hidden units. However, when you show a machine learning expert this graph, they suggest that using 5 or 7 hidden units might be better if the model is to be re-trained. Explain why you think that this suggestion might be made? (4 marks)

(next question over)

Question 3.

Figure 3 is taken from the course textbook:

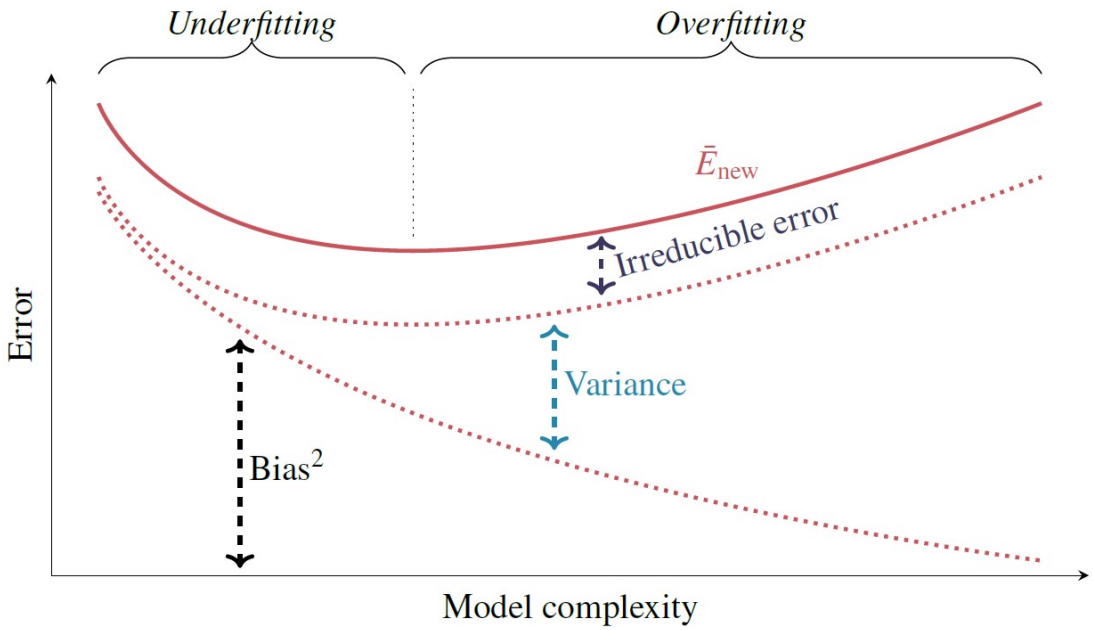

Figure 3

(a) Explain how Figure 3 illustrates a fundamental tradeoff in machine learning. (5 marks)

(question continues over)

(b) If more data becomes available for training, what would be the typical effect on the Bias^2 and Variance curves in Figure 3? (3 marks)

(next question over)

Question 4.

In the course textbook, the following Algorithm is presented:

Algorithm 5.3: Stochastic gradient descent

Input: Objective function $J(\boldsymbol{\theta}) = \frac{1}{n}\sum_{i=1}^{n} L(\mathbf{x}_i, y_i, \boldsymbol{\theta})$, initial $\boldsymbol{\theta}^{(0)}$, learning rate $\gamma^{(t)}$

Result: $\widehat{\boldsymbol{\theta}}$

```
Set t ← 0
while Convergence criteria not met do
    for i = 1, 2, ..., E do
        Randomly shuffle the training data {x_i, y_i}_{i=1}^n
        for j = 1, 2, ..., n/n_b do
            Approximate the gradient using the mini-batch
              {(x_i, y_i)}_{i=(j-1)n_b+1}^{jn_b}, d̂^(t) = (1/n_b) Σ_{i=(j-1)n_b+1}^{jn_b} ∇_θ L(x_i, y_i, θ^(t)).
            Update θ^(t+1) ← θ^(t) − γ^(t) d̂^(t)
            Update t ← t + 1
        end
    end
end
return θ̂ ← θ^(t-1)
```

Figure 4

(a) In this pseudocode, what does $\theta^{(t)}$ represent? (2 marks)

(b) In this pseudocode, what does n represent? (2 marks)

(c) In this pseudocode, what does E represent? (2 marks)

(question continues over)

(d) Explain how the pseudocode in Figure 4 can be simplified if we want to describe (full batch) gradient descent (i.e. if $n_b = n$)? Refer to the line numbers and state which lines of code should be removed or modified in this case. (4 marks)

(next question over)

Question 5.

A fully-connected feedforward neural network can be described mathematically as:

$$\mathbf{q}^{(1)} = h\left(\mathbf{W}^{(1)}\mathbf{x} + \mathbf{b}^{(1)}\right), \tag{1}$$

$$\vdots \tag{2}$$

$$\mathbf{q}^{(L-1)} = h\left(\mathbf{W}^{(L-1)}\mathbf{q}^{(L-2)} + \mathbf{b}^{(L-1)}\right), \tag{3}$$

$$\mathbf{z} = \mathbf{W}^{(L)}\mathbf{q}^{(L-1)} + \mathbf{b}^{(L)}, \tag{4}$$

$$\mathbf{g} = \text{softmax}(\mathbf{z}) = \frac{1}{\sum_{j=1}^{M} e^{z_j}} \begin{bmatrix} e^{z_1} \\ e^{z_2} \\ \vdots \\ e^{z_M} \end{bmatrix} \tag{5}$$

(a) How many layers of weights does this network have? (2 marks)

(b) What does M represent? (2 marks)

(c) What does $h()$ represent? (2 marks)

(question continues over)

(d) Consider a fully-connected feedforward network to be trained on the MNIST dataset, where each input is a 28×28 image. The network is to have two hidden layers, each with 100 units. What is the size (rows $\times$ columns) of the following variables:

(i) $\mathbf{W}^{(1)}$ (2 marks)

(ii) $\mathbf{b}^{(2)}$ (2 marks)

(iii) $\mathbf{W}^{(3)}$ (2 marks)

(next question over)

Question 6.

(a) Given the following matrices:

$$X = \begin{pmatrix} 1 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & 1 \end{pmatrix}, K = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

where X is the 5×5 input data (e.g. pixel values) and K is a 2×2 kernel, perform a convolution operation on X using K (stride of 1).
(Note: only show your working for one kernel position and the complete result). (3 marks)

(b) Perform a convolution operation on X using K but use a stride of 2 and use padding (zero values): add a row to the bottom and a column to the right of X (making it a 6×6 matrix).
(Note: only show your working for one kernel position and the complete result). (3 marks)

(c) Perform 3×3 max. pooling on matrix X (stride of 1). (1 mark)

(next question over)

Question 7.

Figure 5 is from the course textbook and illustrates some properties of a Gaussian Process model for a 1D regression problem. The Gaussian Process uses a squared exponential covariance (kernel) function, with two different values for a hyperparameter. In each case, the model has been trained with the same three data points (shown).

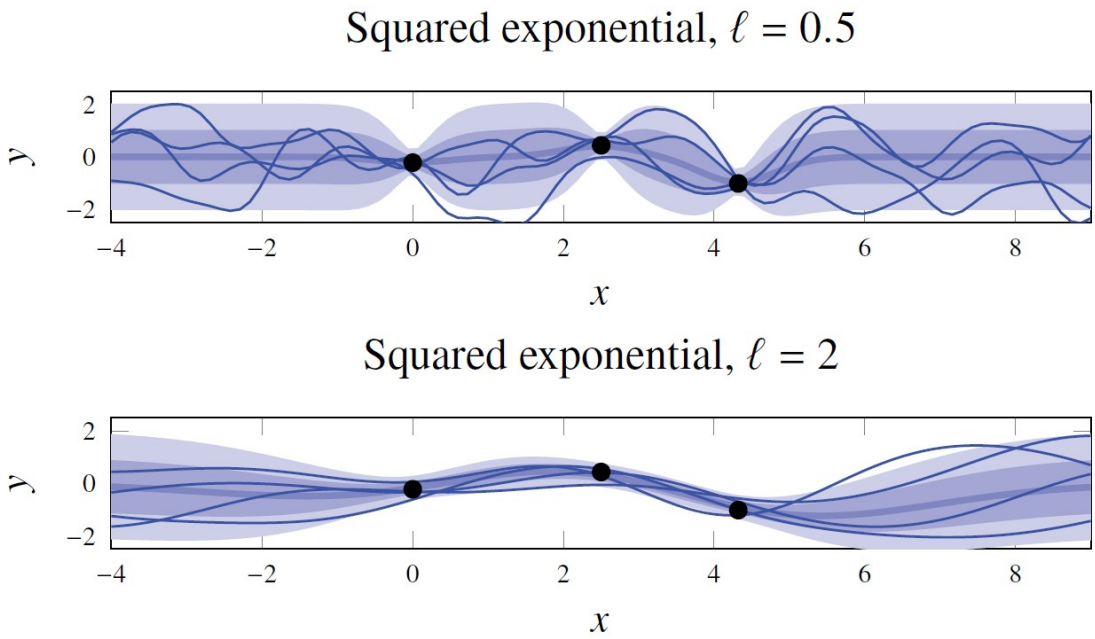

Figure 5

Write a short explanation of what is shown in Figure 5. In particular, explain the shapes of the curves and the shaded regions, and compare the top and bottom sub-figures. (4 marks)

(next question over)

Question 8.

Figure 6 is from the course textbook and shows a 2D Gaussian mixture model that has been trained on the set of data points shown:

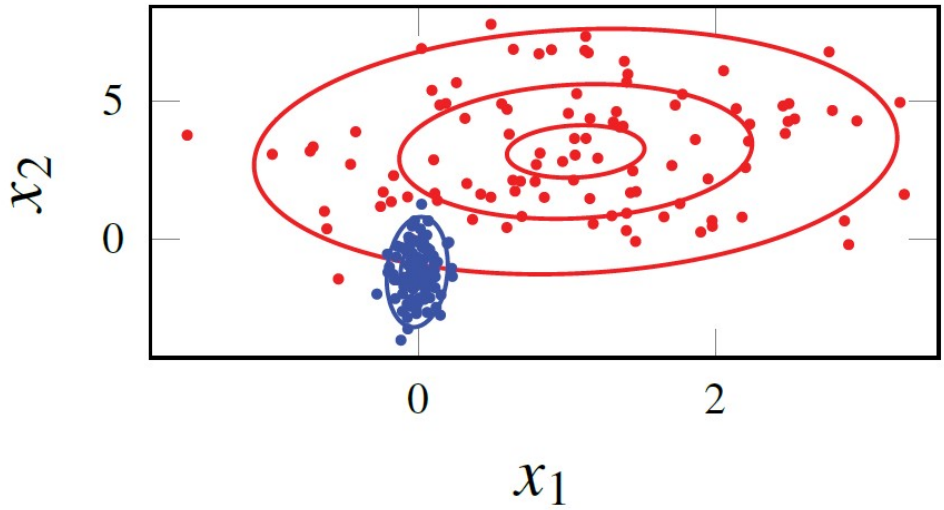

Figure 6

A Gaussian mixture model is an example of a simple generative machine learning model. Write some pseudocode that describes how to generate new data from the mixture model shown in Figure 6. You can define notation to refer to the parameters of the model if you wish, or just use words as long as your steps are clear and precise. (4 marks)

END OF EXAMINATION

This page can be used if you need extra space to answer any of the questions. Clearly indicate the question(s) that you are answering. Ensure that you also put a note within the body of the relevant question(s) directing the marker here.